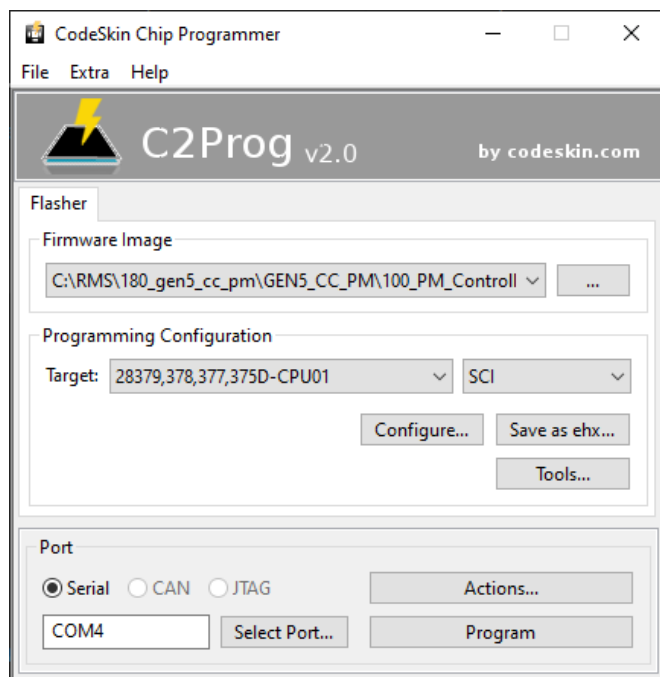


Latest Release:	6533
Firmware Release Type:	Minor Update
Stock EEPROM Release:	20250513
DBC Release:	20250206



Instructions for installing new firmware:

Use latest version of C2Prog from codeskin.com to program the latest firmware as follows. In C2Prog select the latest hex file provided. Select Target 28379,378,377,375D-CPU01. **Note: some early CM200 and CM350SiC units have been built with a target that is 28379,378,377,375S.** These units should have a label indicating that they are a different target. Make sure Smart Sector Selection is selected. Select the serial port that is used to connect to the controller. C2Prog

With the controller powered down, switch the serial connection to program (switch closed), usually via a switch on the serial harness. Power the controller on and hit the program button in C2Prog. When it is finished programming, close the popup window by selecting the OK button. Power the controller down and switch the serial connection to run (switch open).

Stock EEPROMs 20250513		
Unit	Nominal Voltage [Vdc]	STOCK EEPROM
iM-225DX-D	320	05-0031-07a iM-225DX-D Stock EEPROM
iM-225DX-DW	320	05-0037-07a iM-225DX-DW Stock EEPROM
iM-125DX-DW	320	05-0041-07a iM-125DX-DW Stock EEPROM
iM-225DZ-S	650	05-0032-07a iM-225DZ-S Stock EEPROM
iM-225DZ-SW	650	05-0038-07a iM-225DZ-SW Stock EEPROM
iM-375DZ-D	650	05-0033-07a iM-375DZ-D Stock EEPROM
iM-375DZ-DW-SiC	650	05-0147-07a iM-375SiC-DW Stock EEPROM
iM-375DZ-D-SiC	650	05-0039-07a iM-375SiC-D Stock EEPROM
iM-425DZ-D	650	05-0034-07a iM-425DZ-D Stock EEPROM
iM-425DZ-D-SiC	650	05-0040-07a iM-425SiC-D Stock EEPROM
iM-425DZ-D Pump Delete	650	05-0045-07a iM-425DZ-D Pump and HX Delete Stock EEPROM
iM-425DZ-D-SiC Pump Delete	650	05-0046-07a iM-425SiC-D Pump and HX Delete Stock EEPROM
iDM-190	320	05-0035-07a iDM-190DX-D Stock EEPROM
iDM-375	650	05-0036-07a iDM-375SiC-D Stock EEPROM
SS-250-115-DOM + CM200DX	320	05-0031-07a iM-225DX-D Stock EEPROM
SS-250-115-DWM + CM200DX	320	05-0037-07a iM-225DX-DW Stock EEPROM
SS-146-130-DWM + CM200DX	320	05-0041-07a iM-125DX-DW Stock EEPROM
SS-250-115-SOM + CM200DZ	650	05-0032-07a iM-225DZ-S Stock EEPROM
SS-250-115-SWM + CM200DZ	650	05-0038-07a iM-225DZ-SW Stock EEPROM
SS-250-115-DOM + CM350DZ	650	05-0033-07a iM-375DZ-D Stock EEPROM
SS-410-150-DOM + CM350DZ	650	05-0034-07a iM-425DZ-D Stock EEPROM

Please see the “Software Errata” and “Software Feature Additions” documents for more information on bug fixes and feature additions, respectively. Each bug or feature in the changelog below should have a detailed description of impact to the user. These documents can be found in the software release package as of v6533.

GEN5 / CM Firmware Updates			
Change Description	Firmware Revision	Release Date	Date Code
- Fixed an issue with torque rate limiter allowing torque limit violations during inverter state machine transitions. - Fixed an issue with inverter state machine transition interaction with torque rate limiter allowing for longer torque ramp down than transition time allowed. - Fixed an issue with inverter state machine transition which caused torque command to persist through a quick disable-enable transition. - Fixed invalid variable PWM EEPROM write via CAN appearing successful. - Fixed an issue where a variable PWM EEPROM write via CAN could result in EEPROM checksum getting corrupted. - Fixed an issue when a fault occurs that caused inverter state machine oscillations between states 9 and 1. - Fixed Exc_Cmd_Filters defsyms providing incorrect feedback via serial. - Fixed an issue with shudder compensation not obeying torque capability limitations during reverse direction command. - Fixed an issue with torque command Iq/I_d table lookup violating EEPROM motoring and regen torque limits.	6533	05-13-2025	20250513

- Fixed an issue with stall burst feature which would allow excessive current by using higher of coolant temperature or 45c as base for calculation. - Updated inverter hot spot models which commonize implementations and allow for usage with torque capability calculations. Consult the Software User Manual for more information. - Added motor hot spot calculation, replacing stall torque temperature factor for motor protection during stall operation. Consult the Software User Manual for more information. - Added hardware support for CM350SiC @ 500A inverters. - Added active short circuit feature, allowing grounding of motor phases to protect the inverter during certain scenarios. Consult the Software User Manual for more information. Note this also adds three EEPROM values to the defsyms. - Added torque shudder mode 2. Consult the Software User Manual for more information. - Added DC Link field weakening regulator for select motors. Consult the Software User Manual for more information. Note this also adds a EEPROM value to the defsyms. - Added motor temperature EEPROM derate limiting flag to CAN. - Fixed torque shudder allowing more torque than EEPROM max torque. - Fixed torque capability calculations, considering both motor and inverter hot spot current limits. - Fixed regen torque capability calculation incorrectly using a motor torque limit. - Replaced INV_Stall_Burst_Model_Temp signal in CAN DBC with INV_Hot_Spot_Temp_Motor.	6532	02-06-2025	20250206
Added motor type 299 for iM-425x Pump and HX Delete configuration.	6531	08-29-2024	20240829
- Added iM-125DX-DW Support. - Removed stall factor limit over-temperature fault.	6530	05-24-2024	20240524

Updated CM350SiC support.	652F	04-03-2024	20240403
- Re-tuned hot spot estimators for CM200DX. - Improved stall performance for CM200DX. - Fixed torque capability CAN broadcast message. - Separated torque capability CAN signal into motor/regen signals. - Fixed BMS limiting CAN broadcast flag. - Separated BMS limiting flag into motor/regen flags.	652E	03-20-2024	20240320
- Added resolver calibration to inverter calibration backup. - Added CAN access to inverter calibration backup/restore. - Added torque capability CAN broadcast message. - Fixed CM350SiC active discharge. - Fixed CAN broadcast message, temperature 2, being sent at incorrect interval. - Fixed torque request motor/regen processing.	652D	11-17-2023	20231117
Changed inverter calibration backup to be a factory calibration restore feature.	652C	08-29-2023	20230829
- Added inverter calibration backup. - Updated CM350SiC support. - Added support for CM200 hardware update. - Added support for CM200 LIN support. - Fixed CM200/CM350 resolver status not reading correctly. - Updated HVH250-115S motor type gains.	652B	08-23-2023	20230823
Fixed CM350EVO resolver read issue, affecting v6529.	652A	06-14-2023	20230614

• Added support for active discharge on CM350SiC. • Added motor temperature sensor type 44. • Added oil temperature pressure running calibration for iM425. • Fixed race condition between stall burst model and variable PWM stall exit transitions. • Fixed CAN command torque limit not being obeyed during regen. • Fixed regen torque limit check in VSM mode.	6529	06-06-2023	20230606
• Updated CAN EEPROM writing/reading method. • Added missing CAN EEPROM addresses. • Added OBD2 enable CAN command. • Fixed relay 4, 5 statuses not reporting correctly over CAN. • Updated motor type 58 gains.	6528	04-11-2023	20230411
• Updated CM350SiC support. • Improved self-sense assist capability during loss of traction events. • Improved thermal model for stall burst models on all inverters. • Added stall burst models for CM200DZ, CM350DZ. • Re-tuned hot spot estimators for PM100DXR, PM150DX, PM250DZ/DZR, CM200DX, and CM350SiC. • Added motor type 278. • Fixed coolant temperature derate bug. • Improved stall PWM exit performance.	6527	02-28-2023	20230228
• Added continuously variable PWM functionality, removed region-based method. • Fixed issue with active discharge being cancelled when the CAN command has the enable bit set to 0. • Added ability to cancel active discharge by supplying 0 for the request bit. • Fixed issue with current regulator gains. • Re-tuned hot spot estimators for all inverters. • Updated inductance tables for HVH410-150-DOM. • Updated current tables for HVH410-150-DOM. • Fixed incorrect fault code in defsyms file.	6526	12-01-2022	20221201

• Fixed issue with feedback torque filtering over CAN. • Fixed issue with individual Iq/Id limits not being obeyed for IPM motors. • Added hardware support for CM350DZ-EVO. • Removed stall burst model from being limiting factor for PWM frequencies other than 2kHz. • Fixed issue with stall burst model limiting flag false reporting on CAN.	6525	9-27-2022	20220927
• Fixed issue with current regulator limiting too much on bootup. • Fixed issue with temperature based current regulator limiting. • Fixed issue where EEPROM could program when CM GUI reported it did not. • Fixed issue where EEPROM would erase under certain firmware update procedure. • Improved stall burst thermal model tuning. • Updated motor types 47 and 49 to be used for iM-225W. • Added self-sense assist feature to iM-375DZD.	6524	9-13-2022	20220913
• Fixed issue with stall burst thermal model not working correctly at negative speeds. • Fixed issue with variable PWM speed transitions not working correctly. • Added stall burst thermal model flags to CAN.	6523	7-21-2022	20220721
• Added stall burst thermal model to allow short duration higher torque on CM200DX. • Added self-sense assist feature to iM-225DXD to reduce torque oscillation at high speed due to resolver noise. • Improved variable PWM frequency feature. • Fixed an issue with VSM mode not working. Affected versions: 6520, 6521. • Added KL15_EN detection to CM350 power-on self-test. • Improved accuracy of CM350 gate driver board temperatures.	6522	7-17-2022	20220717
• Improved feedback torque filtering and accuracy. • Fixed Ib_corr_fil value being displayed incorrectly in GUI.	6521	5-16-2022	20220516
• Added temperature compensation to stator resistance calculation. • Fixed average module temperature calculation. • Fixed lookup table bug, off-table values were not bounded correctly.	6520	5-13-2022	20220513

• Fixed initialization bug in motor type 16 and 27, incorrect motor inductances being used. • Fixed field weakening PID gain bug, IPM style motors were not using the correct field weakening regulator gain values. • Fixed BMS limit active flag not displaying in GUI. • Improved accuracy of DC bus voltage reading on CM200/CM350. • Reduced noise on CM200/CM350 resolver feedback. • Improved torque accuracy at lower torque ranges of HVH250-115DOM based motors. • Fixed active discharge shutdown bug that could allow torque to build during the inverter disabling process. • Improved fault resistance of CM350 gate driver.	651F	4-18-2022	20220418
• Fixed motor temperature limits. Limits were only applying when receiving a new CAN command. Now limits are applied immediately. • Change how stall current limits are calculated and stall current limit ratio. Now results in higher stall current in some conditions. • Update current limits for CM200DX, CM200DZ, and CM35DZ. • Fix D-axis current limit. Current limit was not set correctly which can result in reduced field weakening performance in some conditions. • Fix PM250DZ current limits, to allow for higher current at low PWM frequency. • Increase Max Torque for motor type 21 and motor type 5 to 600 Nm. • Selectable PWM range is now 2 – 24 kHz. • Added the new motor temp sensor Type 36 for the ring terminal thermistor used on new iM225. • Remove burst current feature. Burst current duration was too short and not needed now that there is variable PWM and higher stall current. • Add a new lower speed higher current region of PWM operation. • Make the stall PWM frequency selectable. • Add in variable stall derate. • Update HVH410-150DOM tables to 900 Arms. • Updated torque limits for iM425 (Motor Type 243). • In circular buffer report PWM frequency instead of 12V sense. • Add ability to trigger a diag. data dump via CAN.	651E	2-10-2022	20220210

• Added ability to disable and reenable shudder compensation via CAN Parameter message. See CAN Protocol manual for more details. • Added ability to send a new maximum speed limit over CAN when in torque mode. Any non-zero value sent in the speed command bytes will result in an update maximum speed limit. • Fixed issue with speed limit function when in Valet mode (VSM mode only). • Motor type 21 now uses w1 for current regulators. • Added current limits for CM350DZ. • Change Ford DU sensors to use 10K external pull-up for better accuracy at low temperature. • New Low Speed High Current and Stall PWM region. • PWM frequency selection down to 2 kHz for CM350. • Fix current regulator settings when using the GUI. • Added temperature sensor type 38 to be reporting for oil pressure/temperature. • Added motor type 243 for iM-425DZ-D configuration. • Updated Iq table for HVH250-115P 650V. • Added Ford Mach E drive unit motor as motor type 241. • Added temperature sensor type 37 for Ford Mach E drive unit temperature sensors. • Added the resolver status registers that hold the sum of the fault conditions to what can be viewed via the GUI.	651D	12-21-2021	20211221
• Add hardware version 37794 for the CM200DZ with new power module.	651C	10-15-21	20211015
• Fixed DC current sensor direction for CM350DZ.	651B	9-24-21	20210924

• Allow Motor Type 64 & 65 to change resolver_rdc_f. • Changed the use of resolver_rdc_f in motor type options EEPROM. Now resolver_rdc_f is used to determine if a resolver is used or if sin/cos encoder is used. If resolver is used then resolver_rdc_f should be 1 and the resolver chip will be used. If a sin/cos encoder is used then resolver_rdc_f should be 0, resolver ship will not be used, and sin/cos ADC inputs are used. Before resolver_rdc_f was being used to determine if resolver chip was to be used for resolver or if separate circuit for resolver was to be used. • Adjust GUI display output for sin_corr and cos_corr to account for the 0.75 gain in the control board circuit. Now 4V input should read -1.5V instead of -1.1V on the GUI. • Allow motor type 65 to do encoder calibration. • Fix display of Sin_Offset_EEPROM and Cos_Offset_EEPROM in GUI when using defsyms_enc_calibration.txt. • Corrected Motor Type EEPROM to only allow the value if it's included in the current group. Otherwise set to previous value. • Increasing the HVH250-115-DOM tables to 900 Arms. • Fix issue with negative speed command when regen fade speed is non zero. • Fix run fault 19 name in GUI. • Replace label Ki_Current_EEPROM_x_10000 with w1_current_EEPROM_(rads/s) in GUI. Ki label was incorrect. Correct label is w1. • Add DC current sensor input to current sensor low post fault and current sensor high post fault. Only used for hardware versions that use a DC current sensor. • Trigger run fault current sensor if DC current sensor input is out of range when inverter is disabled. No longer set dc_cur_sensor_present_f to false if current sensor is not plugged in. Run fault and post fault will indicate that current sensor is not plugged in for DC current sensor on inverters that use a DC current sensor.	651A	9-22-21	20210922
• Fix bug with GEN5 PM inverters. Note GEN5 PM inverters do not work in firmware versions 6517 and 6518. Does not affect CM inverters. • Fix current limit bug at coolant temperature estimate exactly equal to 80°C. Current limit was incorrectly increasing at 80°C. • Fix temperature based current limit for CM200DX between 45°C and 80°C. • Add temperature, voltage and PWM frequency based current limits for CM200DZ.	6519	8-18-21	20210818
• Bug fix related to iM225DZ configuration oil temp sensor.	6518	8-9-21	20210809

• Updated CM200DZ dead-time. • Improved accuracy of the software based DC bus over-voltage detection threshold. • Bug fix related to gate driver diagnostics.	6517	7-19-21	20210719
• Fix test software for production burn-in. • Add BorgWarner HVH146 300Vdc motor as motor type 195. • Add BorgWarner HVH250-115SOM 320Vdc motor with Wuhan Temp Sensor as motor type 20. • Add BorgWarner HVH250-115SOM 650Vdc motor with Wuhan Temp Sensor as motor type 22. • Add BorgWarner HVH250-115SOM 320Vdc motor in iM-225 with oil pressure and temperature sensor as motor type 33. • Add BorgWarner HVH250-115SOM 320Vdc motor in iM-225 with oil pressure and temperature sensor as motor type 34.	6516	6-11-21	20210611
• Adjusted current regulator gains for improved high current/high torque operation of iM225	6515	5-21-21	20210521

• Fix issue with BMS CAN message not being properly recognized. • Modified speed mode torque limits to recognize de-rating limits from speed, BMS current limits and motor temperature. • Modified CAN Status Message (0xAA default ID) to include a bit indicating if the Start Mode has been activated. • Active discharge voltage monitoring changed to use less filtering to reduce delay in transitioning to completed state. • Removed checking for under-voltage fault when in the active discharge completed state. • Added Motor types 205, 209, 212, 215, 219, 220, 224. • Updated resolver and motor characteristics of motor type 73, HVH410-150DOM. • Fixed bug related to a non-zero torque command when using Speed Mode in VSM mode. • Fixed bug related to the torque command being set to zero when in Speed Mode and motor direction is opposite of command. • Fixed bug related to initialization of regen torque limit. • Changed requirement of High Current Speed Limit to be less than 80% of High-Speed Limit (was 70%). • Updated current limits of CM200DX to allow higher currents when coolant temperature is less than 60°C, current limit saturates at 45°C. Maximum current allowed is 700Arms. • Change CAN message signal for Electrical Frequency, if the inverter is in disable state the message signal changes to the Voltage Feedback Speed in rpm.	6514	5-20-21	20210520
• Add EEPROM parameter to control Analog Input Pull-up Control on AIN1 thru AIN3 on CM200. • Increase Gate Over-voltage threshold on CM200. • Updated control table for HVH250-115SOM to allow commands to 500Nm in 650V configuration. • Fix configuration issue with resolver to digital converter on CM200 that can cause erroneous speed readings when the motor is stopped.	6513	3-17-21	20210317
• Fix current sensor gain for PM150DZ GEN5 and PM100DZR GEN5.	6512	3-9-21	20210309

- Modified setup of HVH250-115DOM based motor types to allow commands to 500Nm in the 320V configuration. - Fixed issue related to setting the Max Speed parameter to low values. - Added temperature de-ratings based on coolant temperature estimates. - Added flag in CAN Status message (offset + 0x10) that indicates when coolant temperature derating is occurring. - Turned on auto-restart of CAN bus when it turns off for excessive CAN errors. - Improved CAN diagnostics available via the RMS GUI. - Corrected the CAN priority to match the manual (Priority = 3) when using J1939 option. - Fixed issue with CAN addresses when using J1939 option.	6511	3-3-21	20210303
- Change L100(CM200) gate driver over-voltage fault from a hardware gate fault to an independent fault based on software detection. Over-voltage fault will set bit 26 of the RUN fault double word. - Fixed issue with the RTD selection output on CM200. - Updated RTD default gains for CM200. - Added a new motor type, motor type 25, same as motor type 11 but with the motor direction of rotation swapped. Requires new resolver calibration versus motor type 11.	6510	1-26-21	20210126
- Add support for new revision of L100 control board with new RTD circuit. - Add support for external watchdog timer on L100 control board.	6509	1-11-21	20210111
- Add support for new revision of L100 gate driver board that modifies the module temperature sensor circuit. Previous revision of gate driver board limited module temperature measurement to -20°C. New revision allows down to -40°C. - New revision of L100DX gate driver board requires HW Version be set to 37789.	6508	12-7-20	20201207
- Correct the over-voltage fault threshold for the L100 inverter. Now 500V nominal. - Update DC bus voltage filter to have the same bandwidth as the PWM frequency is changing. - Add capability to send a fast CAN message (at 3ms rate). Message is sent at offset of CAN ID Offset plus 16 when the CAN Active Hi Word is set to 0xFFFFE (low bit is 0). Message contains Torque feedback, Torque Command, Motor Speed, and DC Bus voltage.	6507	12-2-20	20201202

• Update to L100 gate driver diagnostics. • Fix bug related to commanding negative speeds over CAN. • Add flags in CAN Status message for maximum speed limiting, hot spot limiting, and low speed current limiting. • Update current limits of PM150DXR and L100DX inverter. • Correct issues in oil pressure sensor reading for iM-225. • Add ability to change some new EEPROM parameters over CAN.	6506	11-18-20	20201118
• Major Release • Add support for L100DX inverter ◦ Supports programmable pull-up resistors on AIN1, AIN2, AIN3 ◦ Support for new gate driver and resolver to digital converter platform • Add Hot Spot and Coolant Temperature estimators ◦ Add Hot Spot Temperature current limiting feature (set to 125C) ◦ Limit reporting in bit 60 of M170_Internal_States • Add support for iM-225 using HVH250-115DOM core, using Motor Type 21 • Analog Voltage CAN message changes to support oil temperature and pressure • GUI supports reporting of oil temperature and pressure • Correct issues in maximum speed limiting when operating in CAN mode ◦ Max Speed Limiting reported over CAN in bit 59 of M170_Internal_States • Add CAN Rolling Counter feature ◦ To disable feature set CAN_Debounce_Counter_Max_EEPROM to 0. • Low Speed Gain Constant changed to 0.005, same as 2038 software. • Fixed issue with saturation effect initialization for motor type 65 • Corrected issue with torque feedback calculation when using phase advance feature. • Added ability to set the transmission rate of CAN messages (Fast group and Slow group) • Maximum PWM frequency that the software is capable of is now 24kHz ◦ Not all inverters can support this in hardware.	6505	11-9-20	20201109

• Fix bug that caused improper use of math library. This was causing the torque limits to not work correctly. The effects of the bug are very wide and it's not clear what else was affected. Possible areas affected are: torque limits, DC bus voltage frequency output, various measurements, and various reported values. The only known effect was on the CAN motor torque limit and motor temperature de-rate. • Fixed an access invalid index bug. The effects of this bug are unknown. Possible areas affected are EEPROM parameters.	6504	6-1-20	20200601
• Update to support BorgWarner eDM. • Add various new motors from code version 2028. • Add new temperature sensors from code version 2028.	6503	5-14-20	20200514
• Fix max current for hardware version 37760.	6502	3-16-20	20200316
• Fix max current for hardware versions 84 & 85. • Add some fast loop functions to RAM to speed up code.	6501	3-13-20	20200313
• Initial release.	6500	1-10-20	20200110